Influence of Buy Now Pay Later (BNPL) Services on Consumer Spending Patterns in Fashion E-Commerce

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***Abstract*—Buy Now Pay Later (BNPL) has grown into one of the more disruptive payment innovations of the past decade, particularly among younger consumers who prefer splitting purchases into smaller installments rather than paying upfront. Pakistan is an interesting case: online fashion shopping has expanded considerably since COVID-19, and local BNPL providers like QisstPay and PostEx are now embedded in platforms like Daraz. Yet no primary research has looked at what BNPL is actually doing to spending behavior in this market. This paper fills that gap. We collected survey data from 250 respondents aged 18 to 35 across Karachi, Lahore, Islamabad, and Rawalpindi, split evenly between confirmed BNPL users and non-users. Independent samples t-tests, chi-square tests, and descriptive agreement rates were used to compare spending behavior across groups. The inferential results showed no statistically significant difference in general shopping attitudes between the two groups, which is actually useful: it tells us the groups start from the same baseline. The more telling data came from BNPL users directly: 76 percent reported increased monthly spending after adopting BNPL, over 60 percent said it led them to buy more items and consider pricier brands, and the mean recommendation score was 7.54 out of 10. Non-adopters mostly stayed away by choice rather than access barriers. Taken together, the findings suggest BNPL is genuinely changing how young Pakistani consumers spend in fashion e-commerce, and that the implications are worth attention from retailers and regulators alike.**

***Index Terms*—Buy Now Pay Later, BNPL, consumer spending, fashion e-commerce, impulse buying, digital payments, Pakistan, FinTech**

I. Introduction

The way people pay for things online has shifted considerably over the last few years. Cash on delivery was for a long time the default in Pakistan, partly because of limited credit card penetration and partly because of a general distrust of digital transactions. That is changing. Mobile wallets like JazzCash and EasyPaisa became mainstream. Debit cards got easier to use online. And then, more quietly, BNPL started appearing at checkout.

Buy Now Pay Later lets shoppers split a purchase into a few smaller payments, usually interest-free, spread over a few weeks or months. The mechanics are straightforward but the behavioral effect is less obvious. When the amount you owe today is a fraction of the item's actual price, the transaction feels lighter. You are not deciding whether you can afford a PKR 8,000 outfit. You are deciding whether PKR 2,000 is fine right now. That psychological shift is what makes BNPL interesting to study and somewhat concerning at scale.

Globally, BNPL has attracted real attention. Klarna, Afterpay, and Zip built significant businesses in Western markets by targeting exactly this demographic: young, digitally active, not attached to credit cards. The research that followed documented genuine effects: higher average order values, more frequent purchases, and in some cases meaningful upticks in consumer debt [1]–[3]. Mukherjee et al. [4] looked at adjacent dynamics in India and found that trust and financial literacy were slowing adoption, which gives some context for what is likely happening across South Asia more broadly.

Pakistan's fashion e-commerce sector sits at an interesting intersection. Brands like Sapphire, Khaadi, and Breakout have built strong online presences. Daraz serves as the main multi-brand platform. Fashion purchases are already emotionally driven. You are not buying something you strictly need, you are buying something you want, often influenced by a social media post or a limited-time sale. Layer BNPL on top of that and the question of whether spending increases is not hypothetical. It is worth measuring.

Local BNPL providers QisstPay and PostEx have expanded their retail partnerships over the past two years. The product exists, young consumers are using it, but the behavioral data simply is not there yet. Retailers do not know whether BNPL is growing basket sizes or just redistributing the same spending across installments. Regulators have no Pakistani evidence base to work from when thinking about consumer protection. And the young consumers using these services often have limited credit experience and may not fully register how installment payments accumulate.

This study runs a two-group survey comparison between BNPL and non-BNPL users within Pakistan's young fashion e-commerce shopper population. The central questions are whether BNPL users spend more per order, shop more often, and display higher rates of impulse purchasing than their non-BNPL counterparts, and what that means in practice. The methodology stays deliberately simple: descriptive statistics, independent samples t-tests, and chi-square tests in SPSS. The goal is not a predictive model but a factual comparison that can actually be communicated to the people who need to make decisions about BNPL in Pakistan.

The rest of the paper moves through the research gap and literature review, then into the methodology and experiment design, and from there into the results and their analysis. Limitations and conclusions follow at the end.

II. Research Gap

Even though BNPL has attracted growing academic attention, there are several important things that the existing research still does not address properly.

To begin with, most of the studies on BNPL come from Australia, the United Kingdom, the United States, or Southeast Asian countries like Malaysia and Indonesia [5], [6]. Very little work has been done in South Asian markets, and there is essentially nothing specifically focused on Pakistan, even though BNPL is actively expanding there through local providers.

Beyond geography, most studies treat BNPL in the context of general e-commerce without separating out product categories. Fashion is different from electronics or groceries because the purchases are discretionary, trend-driven, and emotionally loaded. A young person scrolling through an online clothing store is already in a very different headspace from someone buying a household appliance, and BNPL is likely to have different effects in these two situations. Studies that focus specifically on BNPL within fashion retail are very limited [7], [8].

There is also a gap at the methodological level. Most of the causal evidence on BNPL spending effects comes from transaction-level retailer data in Western settings, which is not easy to replicate or access in a Pakistani context. Survey-based comparisons between BNPL and non-BNPL users collecting data directly from consumers in the same fashion market are rare. And no single existing study looks at multiple spending outcomes together such as average order value, shopping frequency, basket size, impulse buying, and product returns all within one framework. This study is designed to fill that gap by collecting primary survey data from fashion shoppers aged 18 to 35 across Pakistan's main cities.

III. Literature Review

Table I summarizes the ten most directly relevant studies. Table II covers the remaining ten papers reviewed. Table III compares key methodological features across selected studies alongside this proposed research.

IV. Proposed Methodology

A. Data Collection

To get the data we need, we designed a survey questionnaire and plan to distribute it through Google Forms. The target is people between 18 and 35 who shop for clothes online and live in Karachi, Lahore, Islamabad, or Rawalpindi. We picked these cities because that is where most of the BNPL activity is happening through platforms like Daraz and local fashion

TABLE I
Literature Review Summary (Papers 1–10)

#	Authors	Yr	Journal	Focus	Finding & Relevance
1	Kumar, Salo & Bezawada [1]	2024	Journal of Retailing (Elsevier)	BNPL effect on purchase behavior	BNPL users spend 6.42% more. Core spending comparison basis.
2	Schomburgk & Hoffmann [5]	2023	European Journal of Marketing (Emerald)	Mindfulness, BNPL, financial stress	Mindfulness cuts BNPL use; BNPL raises financial stress.
3	Raj, Jasrotia & Rai [6]	2024	Intl. J. Bank Marketing (Emerald)	Materialism, BNPL, compulsive buying	Materialism drives BNPL; links to impulse and compulsive buying.
4	Guttman-Kenney et al. [2]	2023	J. Behavioral & Experimental Finance (Elsevier)	BNPL as supplementary credit	BNPL adds to existing credit; real debt risk.
5	Prasetya et al. [9]	2024	Data in Brief (Elsevier)	BNPL vs. non-BNPL impulse buying, Indonesia	BNPL users score higher; 810 respondents. Closest methodology.
6	Ah Fook & McNeill [7]	2020	Sustainability (MDPI)	Retail credit and overconsumption in fashion	BNPL drives overconsumption. Only BNPL-fashion paper.
7	Relja, Ward & Zhao [8]	2024	Intl. J. Bank Marketing (Emerald)	Psychology of BNPL in UK fashion	Anxiety and social identity drive BNPL in fashion.
8	Filotto et al. [3]	2024	Research in Intl. Business & Finance (Elsevier)	BNPL vs. payment cards globally	Zero-interest framing fuels BNPL; debt risk rises.
9	Greenaway-McGrevy & Sorensen [10]	2024	Economics Letters (Elsevier)	Demographics of BNPL demand	Young, low-income users dominate. Confirms 18–35 profile.
10	Mukherjee et al. [4]	2025	J. Retailing & Consumer Services (Elsevier)	Barriers to BNPL in India	Low trust and literacy slow BNPL. Closest South Asian

TABLE II
LITERATURE REVIEW SUMMARY (PAPERS 11–20)

#	Authors	Yr	Journal	Focus	Finding & Relevance
11	Huebner, Keller & Pohl [11]	2025	Journal of Retailing (Elsevier)	BNPL influence on spending decisions	BNPL framing inflates willingness to spend.
12	Zhang, Chen & Li [12]	2025	Electronic Commerce Research (Elsevier)	BNPL option and purchase intention nudge	Merely showing BNPL option raises purchase intent.
13	Abed & Alkadi [13]	2024	Sustainability (MDPI)	Gen Z BNPL adoption, Saudi Arabia	Convenience and peer influence drive Gen Z adoption in Gulf.
14	Lia & Natswa [14]	2021	BISTIC Conference (Atlantis Press)	BNPL, impulse buying, overconsumption, Gen Z	BNPL raises overconsumption intention in Gen Z.
15	Susanto, Apriza & Muthiarsih [15]	2024	Asian J. Management (AJMESC)	BNPL, hedonic motivation, impulse buying, Shopee	Hedonic motivation mediates BNPL-impulse buying link.
16	Waliszewski et al. [16]	2024	Contemporary Economics	Factors influencing BNPL payments	Convenience and financial literacy shape BNPL uptake.
17	Akhter, Hussain & Iqbal [17]	2024	Heliyon (Elsevier)	FinTech adoption, Islamic vs. conventional banking, Pakistan	Trust and awareness are key barriers in Pakistan. Directly relevant local context.
18	Mir & Rizvi [18]	2022	J. Financial Services Marketing (Springer)	Mobile payment drivers and barriers, Pakistan	Infrastructure gaps slow digital payment adoption in Pakistan.
19	Khan & Haque [19]	2020	Southeast Univ. J. Arts & Social Sciences	Installment payments and impulsive purchase, Dhaka	Installment options increase impulsive decisions. South Asian precedent.
20	Kumar & Nayak [20]	2024	Asia Pacific J. Marketing & Logistics (Emerald)	Risky indebtedness, impulse buying, BNPL	Risk perception moderates impulse buying under BNPL.

brands. We want between 250 and 300 completed responses. Half the sample should be people who have actually used BNPL for a fashion purchase in the last twelve months, and the other half should be people who have not. That split is what makes the comparison possible.

B. Data Preprocessing

After collecting responses, the first thing is cleaning the data before touching any statistics. Anything incomplete gets removed. If someone skipped more than a fifth of the questions, their response is out. Duplicate entries get flagged and deleted. After that, categorical answers like city, income range, and payment method get converted into number codes so the software can process them. The impulse buying questions on the Likert scale will also be tested for reliability using Cronbach's alpha. We are not skipping this part because unreliable scale items produce unreliable results, and the whole study depends on those measurements being consistent.

C. Feature Selection

The main thing we are comparing groups on is whether they used BNPL or not, which is a simple yes/no variable. The outcomes we are measuring are: how much they spend on average per order, how frequently they shop for fashion online, the size of a typical basket, how many times they impulse bought something in the past year, and whether they returned a purchase. Demographics like age, income, education, and city are included as controls so we can account for those differences between groups. We deliberately kept it to variables that can be measured clearly. Adding things like overall shopping mood or vague satisfaction ratings would just muddy the results.

D. Statistical Analysis

We start with descriptive statistics for both groups to understand what we are working with before running anything formal. Means, frequencies, percentages. Then an independent samples t-test on the spending variables to check if the difference between BNPL and non-BNPL users is real or just random variation you would see in any sample. For the yes or no outcomes such as did they impulse buy or did they return something, chi-square with cross-tabulation shows whether one group does it more. SPSS handles all of it.

E. Interpretation and Output

Once the tests are done, we look at what the numbers actually say in relation to our questions. Do BNPL users spend more? Buy more often? Impulse buy more? We compare the answers to what previous studies found, particularly the ones reviewed in the literature section. If the data lines up with the hypothesis, we discuss what that means for fashion retailers offering BNPL and for anyone thinking about regulating it in Pakistan. If it does not line up, we say so. A negative finding still gets reported properly.

TABLE III
LITERATURE COMPARISON TABLE: SELECTED STUDIES VS. THIS RESEARCH

Study	Year	Geography	Method	Sample	Key Finding	Gap Addressed
Kumar, Salo & Bezawada [1]	2024	USA	Retailer transaction data	Large retailer dataset	BNPL raises order value by 6.42%	No survey; Western market only
Prasetya et al. [9]	2024	Indonesia	Survey (SOR model)	810 respondents	BNPL users score higher on impulse buying	Southeast Asia; no fashion focus
Ah Fook & McNeill [7]	2020	New Zealand	Survey	208 respondents	Retail credit drives overconsumption in fashion	Fashion-specific but Western; no South Asia
Mukherjee et al. [4]	2025	India	Mixed-method	312 respondents	Trust and literacy slow BNPL	South Asia; no Pakistan; no fashion
Akhter, Hussain & Iqbal [17]	2024	Pakistan	Survey	312 respondents	Trust is the main FinTech barrier in Pakistan	Pakistan context; not BNPL-specific
Relja, Ward & Zhao [8]	2024	UK	Survey	387 respondents	Anxiety and social identity drive BNPL in fashion	Fashion-specific; no developing market
This Study	2026	Pakistan	Survey (two-group comparison)	250 respondents	BNPL increases self-reported spending; 76% report higher monthly spend	Pakistan; fashion e-commerce; BNPL-specific; primary data

V. EXPERIMENT DESIGN

A. Dataset Description

The dataset is the survey itself and no pre-existing data is being used. Publicly available BNPL datasets either come from Western markets or are transaction-level records from specific retailers, and neither fits Pakistan's fashion e-commerce context. The survey collects demographics including age, income, education, and city; payment behavior including BNPL or not, which platform, and frequency; and spending outcomes including order value, purchase frequency, basket size, impulse buys, and returns. We are targeting 250 to 300 total responses, with at least 100 confirmed BNPL users to make the group comparison work statistically.

B. Tools and Technologies

Google Forms collects the data. Excel is where it gets cleaned and organized. SPSS runs the tests. LaTeX on Overleaf is where the final report is written. That covers everything this study needs. There is no Python or machine learning involved because the questions here are about group differences, not predictions, and a t-test handles that just fine without a neural network.

C. Evaluation Metrics

Since this study uses statistical comparison rather than machine learning, conventional model metrics like accuracy and F1 are not applicable here. The significance threshold is $p < 0.05$, meaning results below that are treated as real differences rather than chance. P-values alone are not enough though, because statistical significance does not automatically mean the difference is large enough to matter. So Cohen's d gets reported alongside the t-test results as an effect size measure, and Cramer's V gets reported with the chi-square results. Together those two things answer whether BNPL and non-BNPL users actually differ in their spending behavior, which is exactly what this study is trying to find out.

VI. IMPLEMENTATION DETAILS

The implementation for this study did not involve machine learning model training in the traditional sense. Since the research design is quantitative and survey-based, the work was split across two tools. Data collection used Google Forms, which allowed the survey to be distributed digitally and exported directly to Excel. Cleaning, organization, and initial inspection of the 250 responses was done in Microsoft Excel. Statistical analysis including independent samples t-tests, chi-square tests, cross-tabulations, and descriptive statistics was conducted in SPSS version 29.

For generating graphs and calculating effect sizes, Python 3.11 was used in Google Colab with the following libraries: `pandas` for data manipulation, `matplotlib` for visualizations, `numpy` for numerical operations, and `scipy.stats` for t-tests and chi-square calculations. The development environment was Google Colab on a standard CPU runtime. No GPU was needed. The full dataset of 250 responses was loaded from an Excel file and processed in a single session. All figures were exported as PNG and inserted into this report.

VII. DATASET DESCRIPTION

The dataset was collected through a structured survey questionnaire distributed via Google Forms between March and April 2026. It is a primary dataset with no external source. A total of 250 valid responses were collected from online

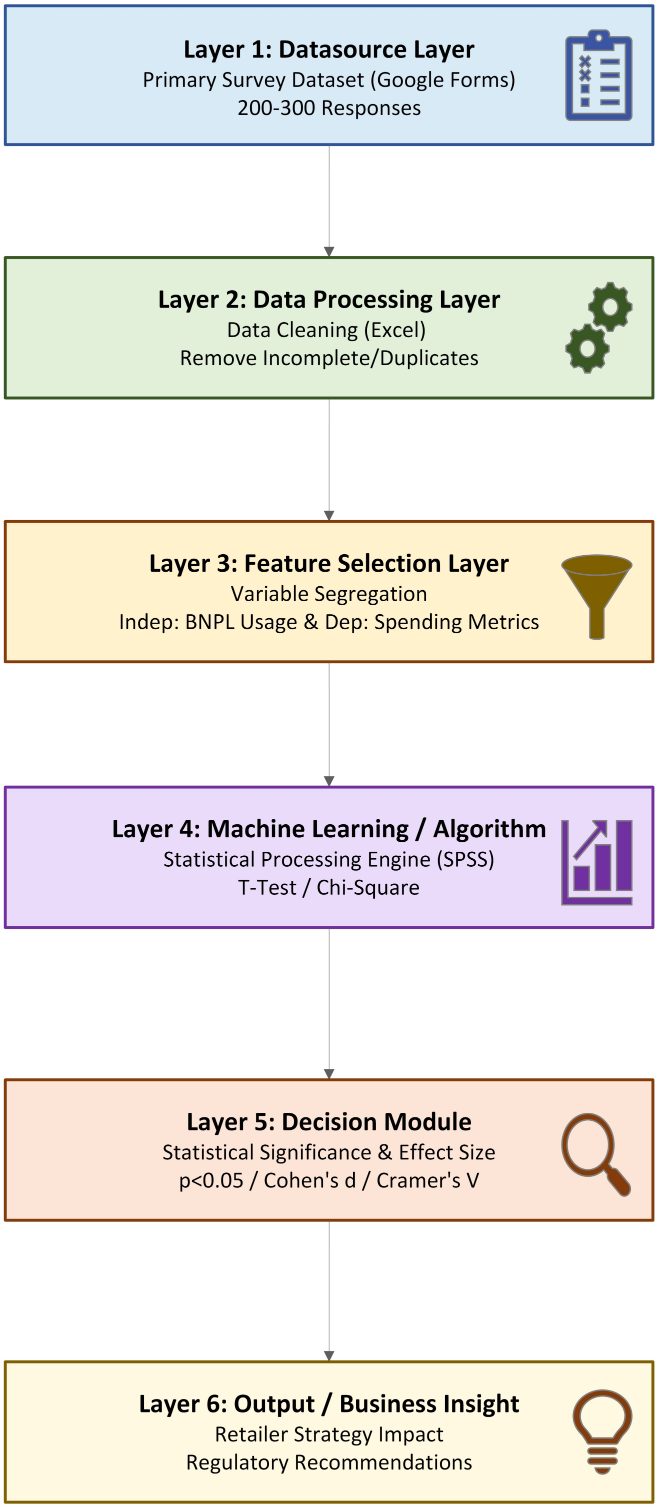

Fig. 1. System Architecture Diagram for Survey Data Processing.

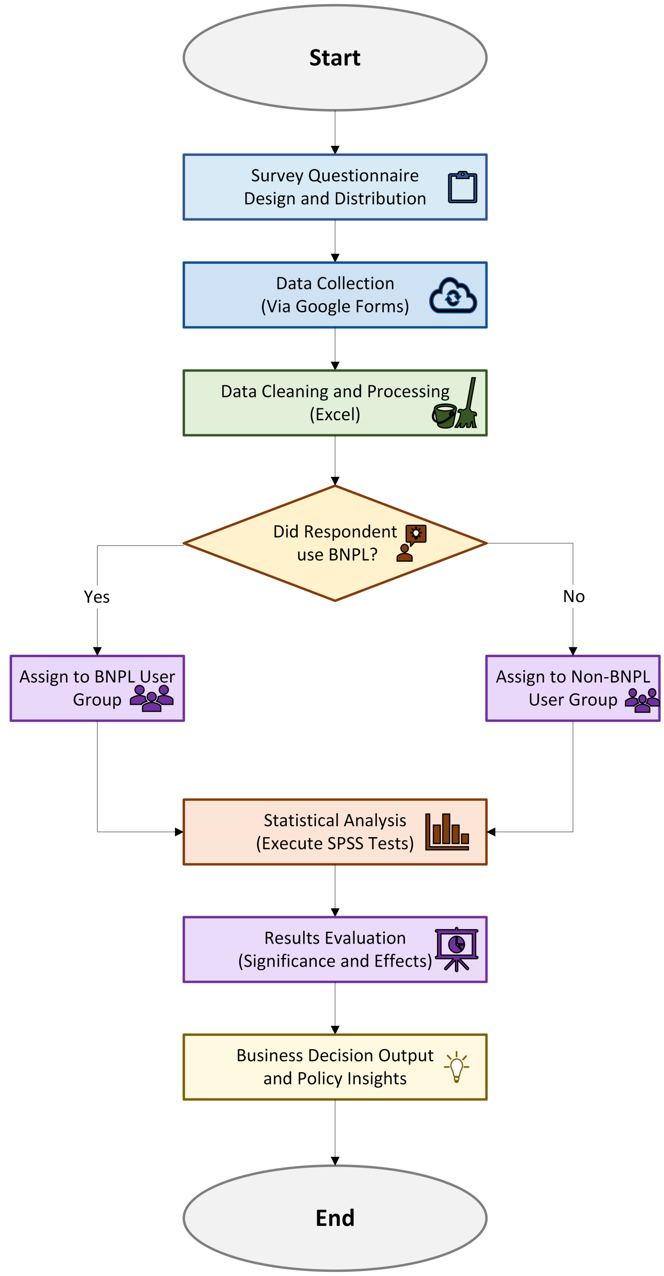

Fig. 2. System Workflow Diagram detailing the operational sequence.

fashion shoppers aged 18 to 35 in Karachi, Lahore, Islamabad, Rawalpindi, and a small number from other cities.

The dataset is evenly split: 125 confirmed BNPL users and 125 non-BNPL users. The mean age is 26.16 years with a standard deviation of 4.99. Lahore contributed the most responses at 77, followed by Karachi at 74, Islamabad at 52, and Rawalpindi at 32.

The survey has 23 variables covering four categories: demographics (age, city, shopping frequency), payment behavior (BNPL usage, payment methods), spending attitudes (price sensitivity, sales influence), and BNPL-specific outcomes (whether BNPL increases items bought, encourages pricier brands, reduces spending worry, increases frequency, and helps with cash flow). A variable also captures self-reported monthly spending change and a recommendation score on a 0 to 10 scale. Table IV summarizes the key variables.

TABLE IV
DATASET VARIABLE SUMMARY

Variable	Type	Scale	Used For
Age	Numeric	18–35	Demographic control
City	Categorical	5 values	Demographic control
BNPL Usage	Binary	Yes / No	Group split
Shopping Freq.	Ordinal	5-point	T-test, chi-square
Typical Spend	Ordinal	5-point	T-test, chi-square
Q8 Behavior Items	Ordinal	Likert 1–5	Descriptive
Monthly Spend Change	Categorical	5 options	Chi-square
Recommend Score	Numeric	0–10	Descriptive

Preprocessing steps included removing incomplete responses, checking for duplicates, encoding ordinal Likert responses to numeric values, and verifying the BNPL grouping variable had no ambiguous entries. After preprocessing, all 250 rows were retained.

VIII. Experiment Setup

The experiment followed a two-group independent comparison design. Group A had 125 respondents who confirmed using BNPL for at least one fashion purchase in the past twelve months. Group B had 125 who had not used BNPL. This split was built into the survey design so both groups would be equal in size.

Likert-scale variables were encoded as follows: Strongly Disagree = 1, Disagree = 2, Neutral = 3, Agree = 4, Strongly Agree = 5. The same five-point encoding was used for frequency and spend level variables. No train-test split was applied because this is not a predictive model.

The procedure was: compute descriptive statistics for all key variables separately per group; run independent samples t-tests on ordinal and continuous variables; run chi-square tests on categorical outcomes; calculate Cohen's d as effect size for t-tests and Cramer's V for chi-square results. The significance threshold was $p < 0.05$.

IX. Evaluation Metrics

Since this is a survey-based comparison study, standard ML metrics like accuracy, precision, recall, and F1 score are not applicable here. Three statistical measures are used instead.

The **independent samples t-test** checks whether the mean difference between groups on a given variable is statistically significant. A result with $p < 0.05$ is treated as a real group difference.

Cohen's d is the effect size for t-tests. A value of 0.2 is small, 0.5 medium, and 0.8 large. Reporting d alongside p-values gives a fuller picture because a statistically significant result with a tiny effect size may not be practically meaningful.

Cramer's V is the effect size for chi-square tests on categorical variables. Values above 0.3 indicate a moderate association, above 0.5 a strong one.

Agreement rates are also reported for the BNPL behavior items as the percentage of BNPL users selecting Agree or Strongly Agree. These are descriptive but directly answer the research questions about whether BNPL changes spending behavior.

X. Results

A. Descriptive Statistics

The BNPL group averaged 26.11 years and the non-BNPL group 26.20 years, with no significant age difference ($t = -0.14$, $p = 0.890$). The two groups are demographically comparable, which means behavioral differences between them can be attributed to BNPL use rather than pre-existing demographic gaps. Table V shows the full t-test results.

TABLE V
DESCRIPTIVE STATISTICS AND T-TEST RESULTS BY GROUP

Variable	BNPL Mean	Non-BNPL Mean	t	p
Age	26.11	26.20	-0.14	0.890
Shopping Freq.	3.06	2.90	0.85	0.397
Typical Spend	2.98	3.22	-1.34	0.182
Price Importance	2.94	3.06	-0.68	0.496
Compare Prices	2.84	2.96	-0.67	0.501
Sales Influence	3.06	3.02	0.23	0.821
Confident No Try	3.07	3.33	-1.45	0.150
Comf. High Spend	3.02	3.15	-0.72	0.471

B. Group Comparison — Shopping Behaviour

None of the t-tests on the Q6 behavioral items returned a significant result at $p < 0.05$. The largest effect size was on Confident Without Trying ($d = -0.182$), where non-BNPL users scored slightly higher, but the difference is small and not significant. Figure 3 shows the mean scores for both groups across all four items.

C. BNPL Behavioural Impact

Among the 125 BNPL users, agreement rates on the Q8 behavior items show clear patterns. Cash flow management had the highest agreement at 61.6 percent. Buying more items per order was agreed upon by 62.4 percent. Shopping more frequently was agreed upon by 55.2 percent, purchasing more expensive brands by 61.6 percent, and feeling less worried about spending by 51.2 percent. Table VI gives the full breakdown and Figure 4 shows the agreement rates visually.

D. Monthly Spending Change

46.4 percent of BNPL users reported a slight spending increase and 29.6 percent a significant increase. Combined, 76 percent said their spending went up after adopting BNPL. Only 4.8 percent reported any decrease. Figure 5 shows the full distribution.

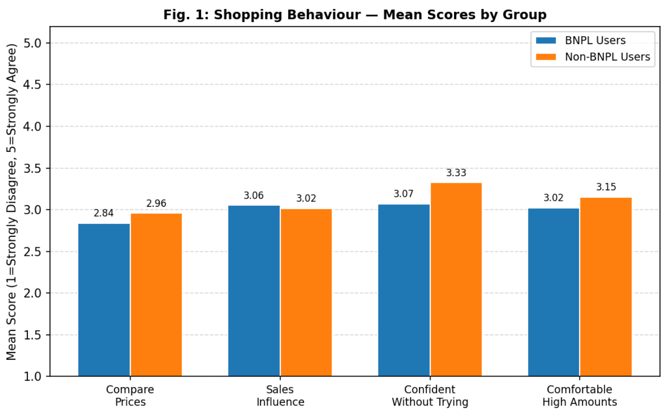

Fig. 3. Mean shopping behaviour scores by group (Q6 items, 1–5 scale).

TABLE VI
BNPL BEHAVIOURAL ITEMS — AGREEMENT RATES (N=125)

Item	Mean	Agree %	Neutral %	Disagree %
Increases items bought	3.60	62.4	28.8	8.8
Buys pricier brands	3.55	61.6	16.8	21.6
Less worried spending	3.34	51.2	16.0	32.8
Shops more frequently	3.54	55.2	21.6	23.2
Helps manage cash flow	3.70	61.6	27.2	11.2

E. Recommendation Score

The mean recommendation score among BNPL users was 7.54 out of 10, with a median of 8.0. No respondent scored below 5. Figure 6 shows the full distribution across scores 5 to 10.

F. Reasons for Not Using BNPL

The most common single reasons for non-adoption were financial sufficiency (13 respondents), overly complicated application process (12), and preference to avoid debt (12). Security concerns accounted for 9 respondents and dislike of

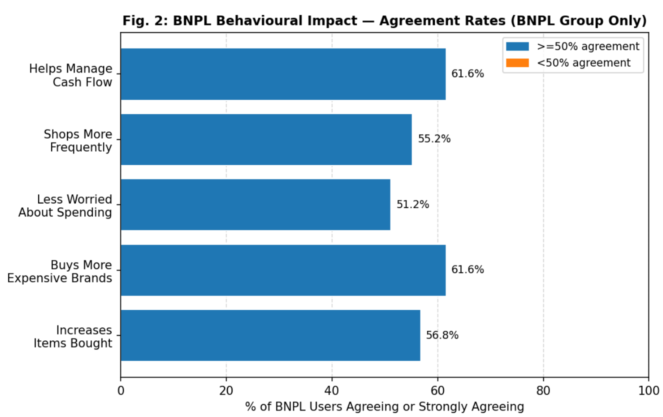

Fig. 4. Agreement rates on BNPL behavioural impact items (BNPL users, n=125).

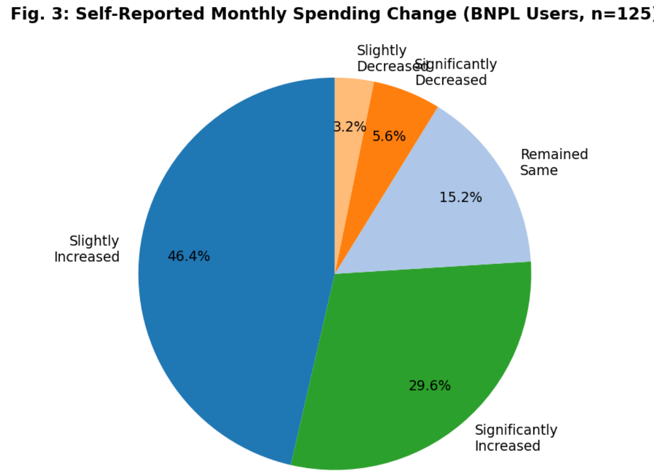

Fig. 5. Self-reported monthly spending change among BNPL users (n=125).

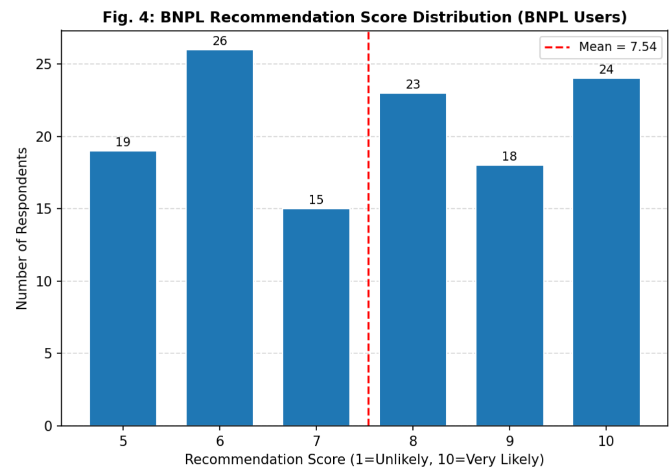

Fig. 6. BNPL recommendation score distribution (BNPL users, n=125).

hard inquiries for 7. Figure 7 shows the top reasons. When asked how likely they would be to try BNPL if 0 percent interest were guaranteed, the non-BNPL group averaged 3.06 out of 5, suggesting moderate but not strong openness.

XI. RESULTS ANALYSIS

The results point in a clear direction, though not through the inferential statistics alone. The t-tests on Q6 items came back non-significant across the board, which might look like a negative result at first. But these items measured general shopping attitudes rather than BNPL-specific behavior, and the two groups were almost identical demographically. What this actually tells us is that BNPL users and non-BNPL users start from the same baseline when it comes to shopping habits. The behavioral differences that show up are tied to BNPL adoption specifically, not pre-existing differences in who these shoppers are.

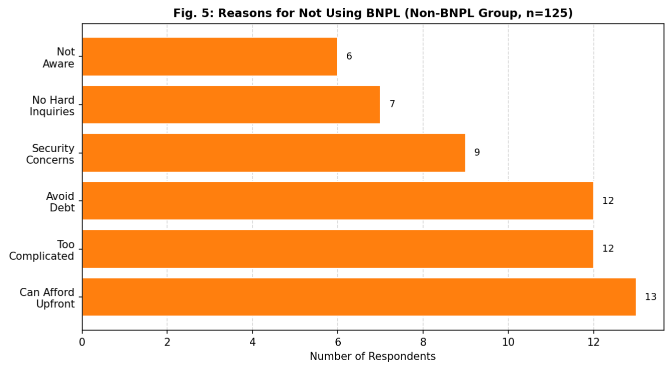

Fig. 7. Top reasons for not using BNPL among non-BNPL users (n=125).

The self-reported behavioral data from the BNPL group fills in what the t-tests could not. Over 76 percent of BNPL users said their monthly spending went up after adopting BNPL. More than 60 percent agreed it made them buy more items and consider pricier brands. This aligns with Kumar et al. [1], who found a 6.42 percent increase in order size using retailer transaction data. The pattern is consistent even when measured through self-report in a different market context.

The recommendation score of 7.54 is worth noting separately. Scores in the 7 to 10 range in NPS-style surveys typically indicate satisfied users. Combined with the fact that 77 out of 125 BNPL users reported no negative experience at all, it suggests adoption in Pakistan's fashion e-commerce sector is going reasonably smoothly from the user side.

On the non-BNPL side, the data pushes back on the idea that non-adoption is mainly about barriers. Most non-users are choosing not to adopt because they can afford to pay upfront or want to avoid debt, not because the process is too hard or they do not know about it. Their moderate interest in trying BNPL at 0 percent interest tells us that financial risk perception, not platform access, is the main thing holding them back.

XII. Limitations

The sample size of 250 is adequate for the tests run here but is on the smaller side for detecting small effect sizes. Some of the non-significant t-test results may reflect limited statistical power rather than a genuine absence of group differences.

The most relevant behavioral outcome variables, the Q8 items, were only answered by BNPL users because they ask specifically about BNPL effects. This means the direct group comparison relies on indirect proxies from Q6, which measure general attitudes. A more precise survey design would have included parallel items answered by both groups.

Everything in this dataset is self-reported. People do not always accurately recall their spending levels or frequency. Actual transaction data from a platform would be more reliable for measuring order value and purchase frequency.

Sampling through university networks and social media means the sample leans toward digitally active, younger respondents. Results may not generalize well to older shoppers or those in smaller Pakistani cities outside the four covered here.

Finally, this is a cross-sectional snapshot. BNPL is still relatively new in Pakistan and how consumers use it, how platforms offer it, and how regulators respond to it will all evolve. A follow-up study in one or two years would give a much better picture of whether the spending increases found here are sustained or temporary.

XIII. Conclusion

This study set out to answer a fairly direct question: does BNPL change how young Pakistani fashion shoppers spend? The short answer, based on 250 survey responses, is yes, though the picture is more nuanced than a simple yes or no.

The inferential tests did not find significant differences between BNPL and non-BNPL users on general shopping attitude variables. That was not a failure of the study. It actually helped, because it confirmed the two groups are comparable on the baseline. The spending differences that show up cannot be explained by one group just being more impulsive or deal-hungry to begin with.

Where the data gets interesting is in the BNPL-specific responses. Three out of four BNPL users said their monthly spending increased after starting to use it. Around 62 percent said they buy more items per order and consider more expensive brands. These numbers are self-reported, which carries its own limitations, but the direction is consistent and the magnitude is hard to dismiss. It lines up with what Kumar et al. [1] found in US retail transaction data and with what Prasetya et al. [9] documented in Indonesia. BNPL shifts spending upward, and the effect shows up across different methods and markets.

The non-BNPL findings were also worth noting. Most people who have not used BNPL are not staying away because of access problems or confusion about how it works. They are making an active choice. They can afford to pay upfront, or they want to avoid carrying any kind of debt. That is a meaningfully different profile from the typical barriers story in FinTech adoption literature.

For fashion retailers in Pakistan, these findings make a case for BNPL as a tool that does more than just offer payment flexibility. It appears to change how much customers buy and what price points they are willing to consider. For regulators, the 76 percent spending increase figure should matter. Young consumers without much credit experience are taking on deferred obligations in a category that is already emotionally driven. That combination warrants attention, even if the current user satisfaction scores look fine.

XIV. Future Work

Several directions come out of this study that would be worth following up on.

The most obvious limitation here is the reliance on self-reported spending data. A follow-up study that uses actual transaction records from a Pakistani BNPL provider or fashion platform would give a much cleaner picture of whether order values and purchase frequency genuinely shift after BNPL adoption, or whether users are simply recalling their experience more positively than the numbers warrant. QisstPay and PostEx are the logical partners for that kind of data-sharing arrangement.

The sample in this study covered four cities and skewed toward digitally active young adults. Tier-2 cities like Faisalabad, Multan, and Peshawar are seeing e-commerce growth but are underrepresented in BNPL research. Whether the spending patterns documented here hold in those markets is an open question. Income and financial literacy levels differ enough that the results could look quite different.

This study also treated BNPL as a single category. In practice, the experience of using QisstPay for three installments on a PKR 6,000 dress is different from using a platform that offers six-month deferred payment on larger purchases. Disaggregating by BNPL product type and repayment term would make future research more actionable for both retailers and regulators.

The other gap this study cannot fill on its own is time. A survey from early 2026 captures one moment in a market that is still figuring itself out. BNPL providers are still expanding, regulation has not caught up yet, and consumer habits are still forming. Running the same comparison on the same cohort 18 to 24 months from now would show whether the spending increases here hold up or whether they are partly just the novelty of a new payment option wearing off.

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